Homework 2

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Physics 16
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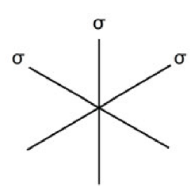
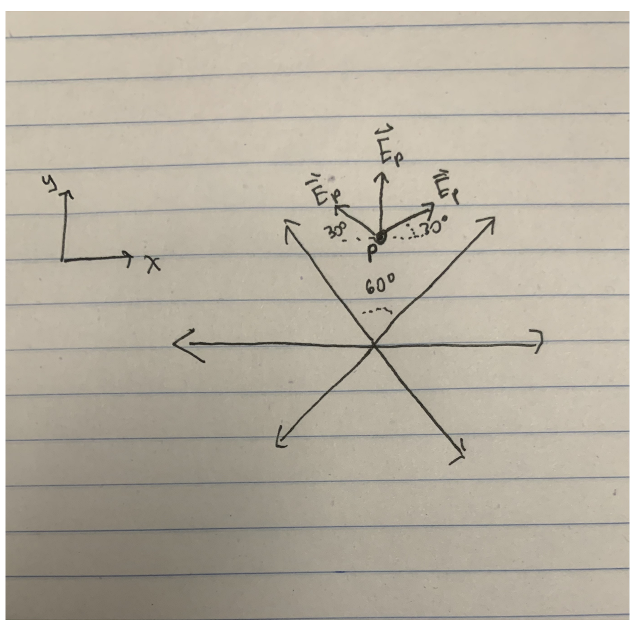
Question 1. The figure shows the cross section of three infinite sheets intersecting at equal angles. The sheets all have the same surface charge density σ. Find the direction and magnitude of the electric field at all points in space.



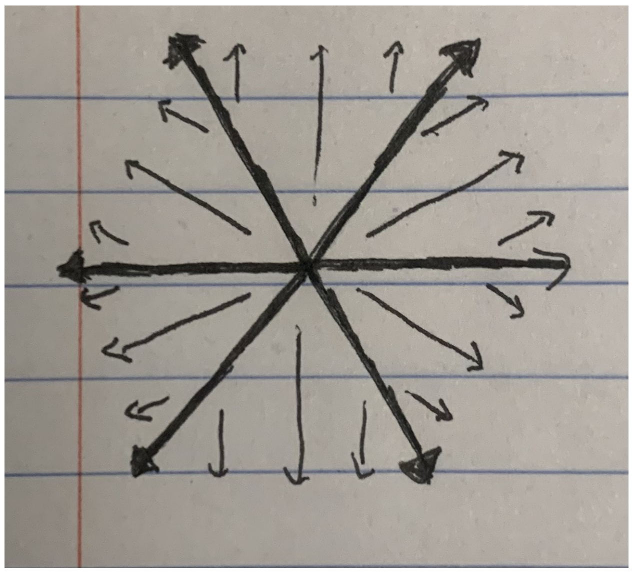
Solution. Suppose we place a point charge p inside of one of the six sections as such:



Recall that a infinite sheet has constant electric field $|E_p| = \frac{\sigma}{2\varepsilon_0}$. By basic vector addition, the two horizontal components cancel out, and hence

$$|E| = 2|E|_p \sin 30^\circ + |E|_p = 2\left(\frac{\sigma}{2\varepsilon_0}\right)\left(\frac{1}{2}\right) + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}.$$

By rotational symmetry, this magnitude of electric field is constant regardless of position. Therefore, $\boxed{|E| = \frac{\sigma}{\varepsilon_0} \text{ everywhere.}}$ We can draw fields lines as shown below to show directionality.



$\square$

Question 2. (Purcell ch. 1, ex. 1.61 — *Potential energy of a sphere.*) A spherical volume of radius R is filled with charge of uniform density ρ. We want to know the potential energy U of this sphere of charge, that is, the work done in assembling it. The result from Purcell Section 1.15 is

$$U = \frac{3Q^2}{20\pi\varepsilon_0 R},$$

calculated by integrating the energy density of the electric field. Derive U here by building up the sphere layer by layer, making use of the fact that the field outside a spherical distribution of charge is the same as if all the charge were at the center.

Solution. Suppose we have built a sphere of radius r where $0 \leq r \leq R$ and the charge $q(r)$ currently contained within this sphere is given by $\rho \cdot \frac{4}{3}\pi r^3$. We now bring an infinitesimal shell of thickness dr from infinity with a charge dq defined as $\rho \cdot (4\pi r^2 dr)$. The work dU required to bring dq to the surface of the existing sphere is $dU = V(r)dq$ and since the field outside the sphere is equivalent to that of a point charge at the center the potential at the surface is

$$V(r) = \frac{1}{4\pi\varepsilon_0}\frac{q(r)}{r}.$$

Substituting our expressions for charge and potential yields

$$\begin{aligned} dU &= \left(\frac{1}{4\pi\varepsilon_0}\frac{\rho\frac{4}{3}\pi r^3}{r}\right)\left(\rho 4\pi r^2 dr\right) \\ &= \frac{4\pi\rho^2}{3\varepsilon_0} r^4 dr. \end{aligned}$$

Now integrate from $r = 0$ to $r = R$ which results in

$$U = \int_0^R \frac{4\pi\rho^2}{3\varepsilon_0} r^4 dr = \frac{4\pi\rho^2}{3\varepsilon_0}\left[\frac{r^5}{5}\right]_0^R = \frac{4\pi\rho^2 R^5}{15\varepsilon_0}.$$

We must now substitute back the total charge is $Q = \rho\frac{4}{3}\pi R^3$. We can solve for ρ to get the square $\rho^2 = \frac{9Q^2}{16\pi^2 R^6}$

substituting this back into the expression for U gives

$$U = \frac{4\pi R^5}{15\varepsilon_0}\left(\frac{9Q^2}{16\pi^2 R^6}\right) = \frac{36\pi Q^2 R^5}{240\pi^2\varepsilon_0 R^6} = \boxed{\frac{3Q^2}{20\pi\varepsilon_0 R}.}$$

□

Question 3. (Purcell ch. 1, ex. 1.62 — *Electron self-energy.*) At the beginning of the twentieth century the idea that the rest mass of the electron might have a purely electrical origin was very attractive, especially when the equivalence of energy and mass was revealed by special relativity. Imagine the electron as a ball of charge of constant volume density out to some maximum radius r_0. Using the result of Problem 2, set the potential energy of this system equal to mc^2 and solve for r_0. One defect of the model is rather obvious: nothing is provided to hold the charge together!

Solution. While the result of Problem 2 directly gives us U, let us find U in a different approach. Divide the field E into two sections, E_{in} and E_{out}, each evaluated through Gauss' Law. All the fields are rotationally symmetric, so we can treat field as a constant multiplying it by the surface area of the sphere. For the inner field,

$$E_{\text{in}} \int dA = \frac{Q_{\text{inc}}}{\varepsilon_0} \implies E_{\text{in}}(\cancel{4\pi} r^2) = \frac{\rho\left(\frac{\cancel{4}}{3}\cancel{\pi} r^3\right)}{\varepsilon_0} \implies E_{\text{in}} = \frac{\rho r}{3\varepsilon_0}.$$

For the outer field, the charge is no longer dependent on r, so

$$E_{\text{out}} \int dA = \frac{Q_{\text{inc}}}{\varepsilon_0} \implies E_{\text{out}}(\cancel{4\pi} r^2) = \frac{\rho\left(\frac{\cancel{4}}{3}\cancel{\pi} r_0^3\right)}{\varepsilon_0} \implies E_{\text{out}} = \frac{\rho r_0^3}{3\varepsilon_0 r^2}.$$

With both these electric fields, we get

$$\begin{aligned}
U = \frac{\varepsilon_0}{2}\int_\nu E^2 d\tau &= \frac{\varepsilon_0}{2}\left[\int_0^{r_0} E_{\text{in}}^2 d\tau + \int_{r_0}^{\infty} E_{\text{out}}^2 d\tau\right] \\
&= \frac{\varepsilon_0}{2}\left[\int_0^{r_0}\left(\frac{\rho r}{3\varepsilon_0}\right)^2 4\pi r^2 dr + \int_{r_0}^{\infty}\left(\frac{\rho r_0^3}{3\varepsilon_0 r^2}\right)^2 4\pi r^2 dr\right] \\
&= \frac{\varepsilon_0}{2}\left[\frac{4\pi\rho^2}{9\varepsilon_0^2}\int_0^{r_0} r^4 dr + \frac{4\pi\rho^2 r_0^6}{9\varepsilon_0^2}\int_{r_0}^{\infty}\frac{dr}{r^2}\right] \\
&= \frac{\varepsilon_0}{2}\left[\frac{4\pi\rho^2}{9\varepsilon_0^2}\left(\frac{r_0^5}{5}\right) + \frac{4\pi\rho^2 r_0^6}{9\varepsilon_0^2}\left(\frac{1}{r_0}\right)\right] \\
&= \frac{1}{2}\left[\frac{4\pi\rho^2}{9\varepsilon_0}\left(\frac{6r_0^5}{5}\right)\right] = \boxed{\frac{4\pi\rho^2 r_0^5}{15\varepsilon_0}.}
\end{aligned}$$

Finally,

$$\frac{4\pi\rho^2 r_0^5}{15\varepsilon_0} = mc^2 \implies \boxed{r_0 = \left(\frac{15\varepsilon_0 mc^2}{4\pi\rho^2}\right)^{1/5}.}$$

□

Question 4. (Purcell ch. 1, ex. 1.59 — *Zero field inside a cylindrical shell.*) Consider a distribution of charge in the form of a hollow circular cylinder, like a long charged pipe. In the spirit of Purcell problem 1.17, show that the electric field inside the pipe is zero.

Solution. Consider an arbitrary interior point P. Let us draw two intersecting lines across P to create two sections within the cross section circle of the pipe. This creates two opposing arc elements on opposite sides of P, at distances r_1 and r_2. Each arc element spans an arc length $r\,d\theta$ along the circumference of the cylinder, and since the cylinder is infinite in length, each element constitutes an infinite line charge with linear charge density

$$d\lambda \propto \sigma r\,d\theta.$$

The electric field at P due to an infinite line charge at distance r is

$$dE = \frac{d\lambda}{2\pi\varepsilon_0 r}.$$

Substituting,

$$dE_1 \propto \frac{\sigma r_1\,d\theta}{2\pi\varepsilon_0\,r_1} = \frac{\sigma\,d\theta}{2\pi\varepsilon_0}, \qquad dE_2 \propto \frac{\sigma r_2\,d\theta}{2\pi\varepsilon_0\,r_2} = \frac{\sigma\,d\theta}{2\pi\varepsilon_0}.$$

Since both E_1 and E_2 are in opposite directions, the integration cancels both sides out so that the electric field is zero at all interior points. $\square$

Question 5. (Purcell ch. 2, ex. 2.46) The right triangle shown in Fig. 2.48 of Purcell & Morin has vertex P at the origin, base b, altitude a, and uniform surface charge density σ. Determine the electric potential at the vertex P. First find the contribution of the vertical strip of width dx at position x. Show that the potential at P can be written as

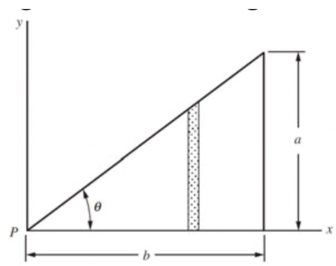


Figure 2.48. (Purcell & Morin)

$$\phi_P = \frac{\sigma b}{4\pi\varepsilon_0}\ln\left(\frac{1+\sin\theta}{\cos\theta}\right),$$

where θ is the angle at P.

Solution. We seek to find a double integral using the general form

$$\phi = \frac{1}{4\pi\varepsilon_0}\int \frac{\sigma}{r}dA$$

Notice that cartesian coordinates will work, where when x changes from 0 to b, y is restricted from 0 to $x\tan\theta$. Hence,

$$\phi = \frac{1}{4\pi\varepsilon_0}\int_0^b\int_0^{x\tan\theta}\frac{\sigma}{\sqrt{x^2+y^2}}dydx.$$

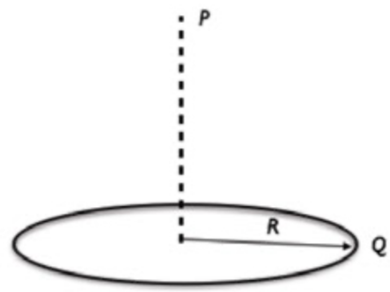
Solving the inner integral requires a substitution technique. Let $y = x\tan\theta'$. Then $dy = x\sec^2\theta' d\theta'$, reesulting in

$$\phi = \frac{\sigma}{4\pi\varepsilon_0}\int_0^b\int_0^\theta\frac{\sec^2\theta'}{\sec\theta'}d\theta' dx = \frac{\sigma}{4\pi\varepsilon_0}\int_0^b dx[\ln|\sec\theta'+\tan\theta'|]_0^\theta$$

$$= \frac{\sigma b}{4\pi\varepsilon_0}\ln\left(\frac{1}{\cos\theta}+\frac{\sin\theta}{\cos\theta}\right) = \boxed{\frac{\sigma b}{4\pi\varepsilon_0}\ln\left(\frac{1+\sin\theta}{\cos\theta}\right)}.$$

$\square$

Question 6. Consider a ring with total charge Q and radius R. Let point P be a distance z from the plane of the ring, along the axis through its center.



(a) By adding up the contributions from all the pieces of the ring, find the electric potential at point P on the axis.

(b) Use your result from part (a) to calculate the electric field at point P on the axis.

(c) A charge $-q$ with mass m is released from rest far away along the z-axis. What is its speed when it passes through the center of the ring (the $z = 0$ plane)? Assume the ring is fixed in place.

Solution. For part (a), we construct a simple integral as such

$$\phi = \int \frac{1}{4\pi\varepsilon_0}\frac{dq}{r} = \int_0^Q \frac{1}{4\pi\varepsilon_0}\frac{dq}{\sqrt{R^2+z^2}} = \boxed{\frac{1}{4\pi\varepsilon_0}\frac{Q}{\sqrt{R^2+z^2}}.}$$

For part (b), we use the gradient to calculate $\vec{E}$. Since, the potential is only dependent of the z variable in cartesian, we can simply find the partial only with respect to z so that

$$\vec{E} = -\nabla\phi = -\frac{Q}{4\pi\varepsilon_0}\frac{\partial}{\partial z}\left(R^2+z^2\right)^{-1/2}\hat{z} = \boxed{\frac{1}{4\pi\varepsilon_0}\left(\frac{Qz}{(R^2+z^2)^{3/2}}\right)\hat{z}.}$$

For part (c), this requires a simple conservation of energy. From an infintisimal distance ($r \to \infty$) to the center of the ring ($z = 0$), the change in potential is

$$\Delta U = -q\Delta\phi = -\frac{Qq}{4\pi\varepsilon_0}\left(\frac{1}{R} - \lim_{r\to\infty}\frac{1}{r}\right) = -\frac{1}{4\pi\varepsilon_0}\frac{Qq}{R}.$$

Therefore,

$$\Delta K = -\Delta U \implies \frac{1}{2}mv^2 = \frac{1}{4\pi\varepsilon_0}\frac{Qq}{R} \implies \boxed{v = \sqrt{\frac{1}{4\pi\varepsilon_0}\frac{2Qq}{mR}}.}$$

$\square$